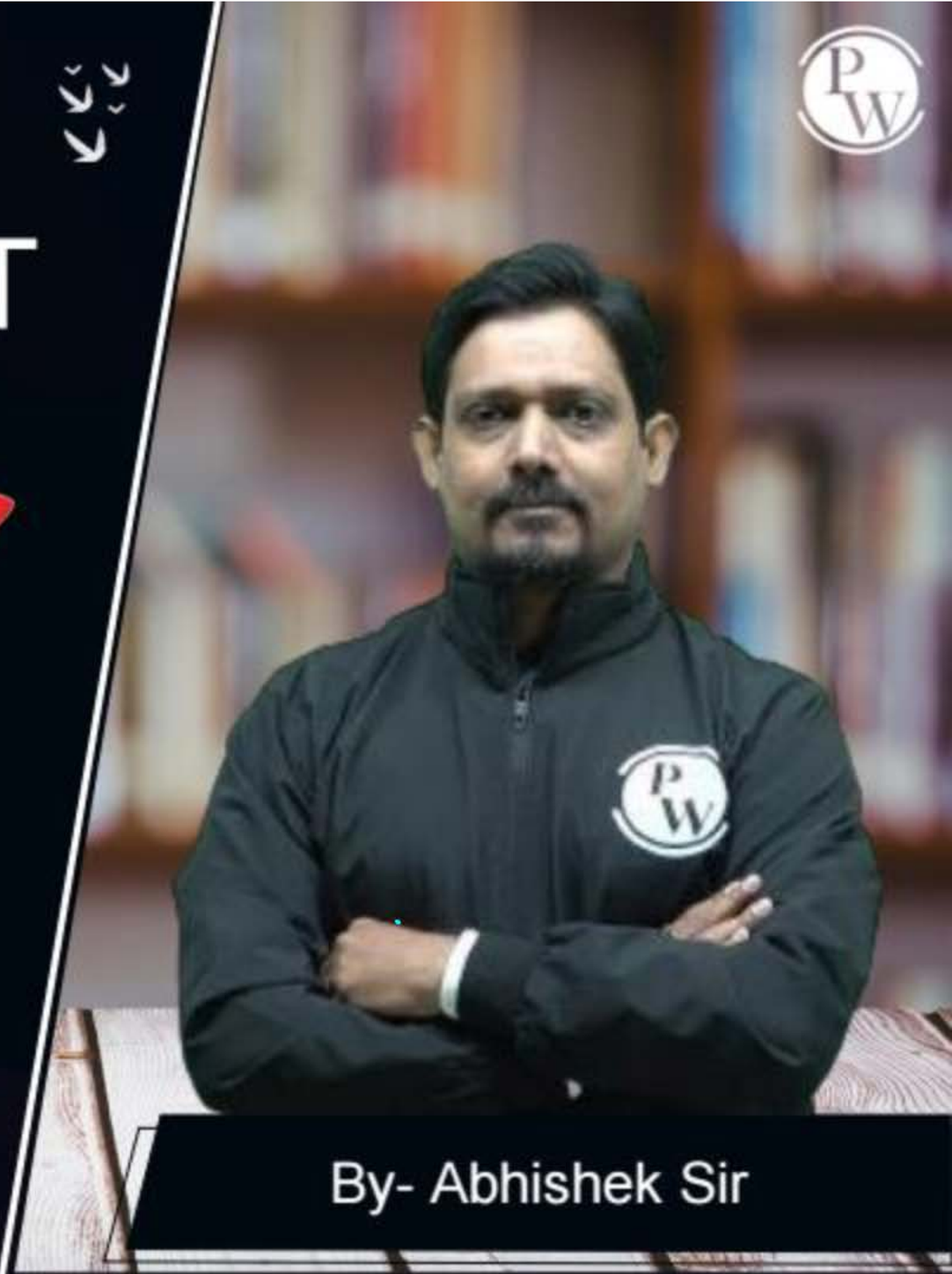
# Computer Science & IT



## C Programming

## Structure & Union

Lecture No. 01



By- Abhishek Sir

# Recap of Previous Lecture



- Topic — String function
- Topic — Structure
- Topic
- Topic
- Topic

∴ dot operator

Slide

# Topics to be Covered



- Topic — Structure
- Topic — Union
- Topic
- Topic
- Topic

3 gate question

Slide



# Structure

During declaration strucue variable cant be initlizlize
Structure pointer
Arrow operator
Structure pointer
Typedef
Union



# Structure

```
struct A {
    int i;
    char ch;
};
```

Structure is collection of dissimilar data type

Member of structure

```
struct A var = { 10, 'c' };
```

Initialization of structure variable.



# Structure

if we want to access member of structure then we use '.' operator

```
printf("%d", var.i); // 10 print
printf("%c", var.ch); 'c' character print
```



# Structure

Every time structure variable to declare we Need to write

Struct A

typedef struct S {

char *name;

int RollNo;

} student;

typedef creates

alias : another

Common

general Name.

Slide



# Structure

struct S. $S_1$;

Student $S_1$; ← readable and it carries more Sense

typedef alias. struct S



# Structure

**Structure pointer** :- Structure pointer will hold address of structure variable

```
Struct s s1 = {"vijay", 23};

Struct s *ptr
Student *ptr;
Student *ptr = &s1;
```

← Inchlizahm

Slide



# Structure

declaration : ptr = &S1;

How to access member of a structure using structure pointer.

"→" arrow operator.

↑ dereference operator

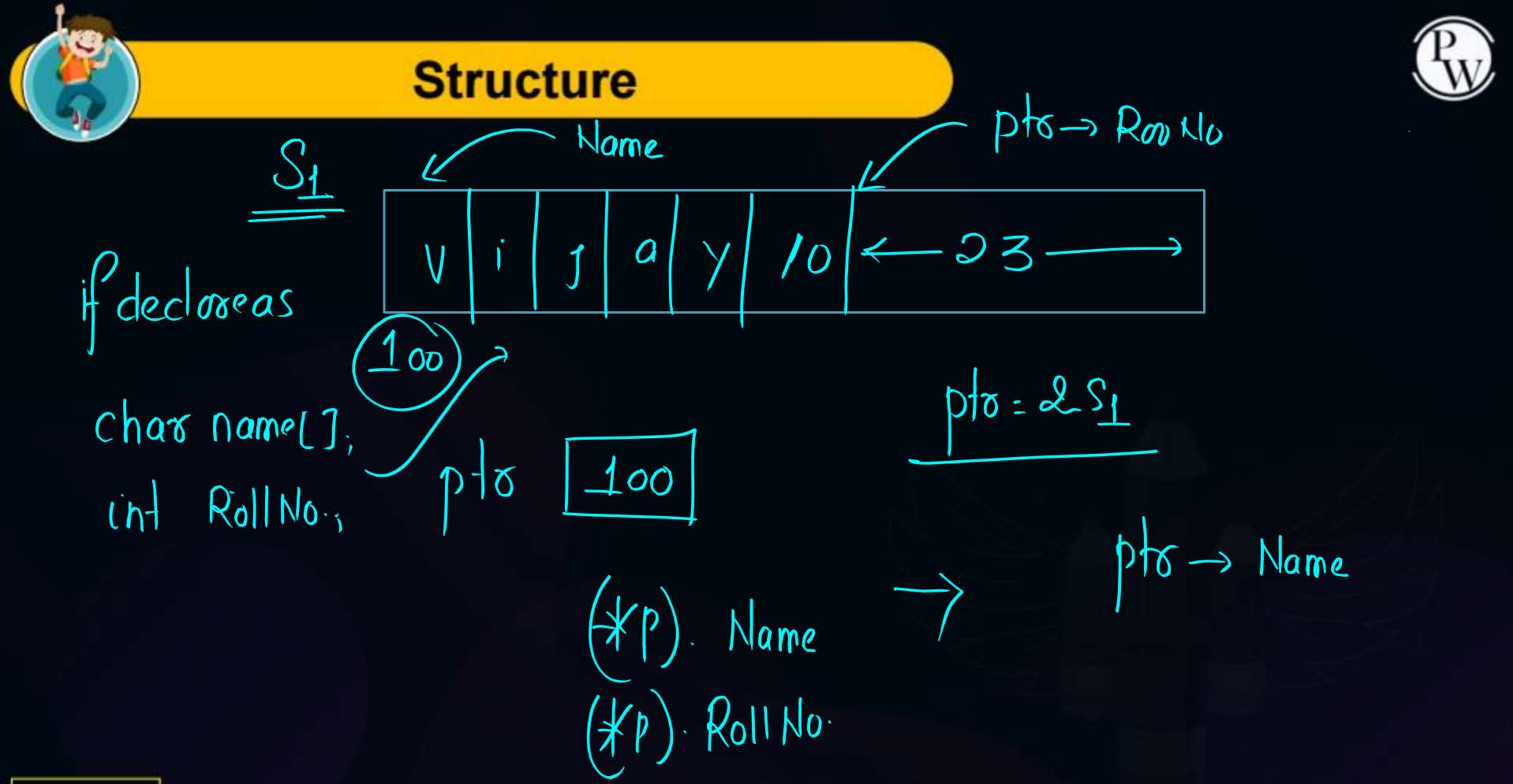
printf("%s", ptr→name);

printf("%d", ptr→rollno);

printf("%d", (*p).roll No);

Slide

# Structure




Slide



# Structure

typedef struct name{

alias

} Name;

struct name var;

Name var;

Slide

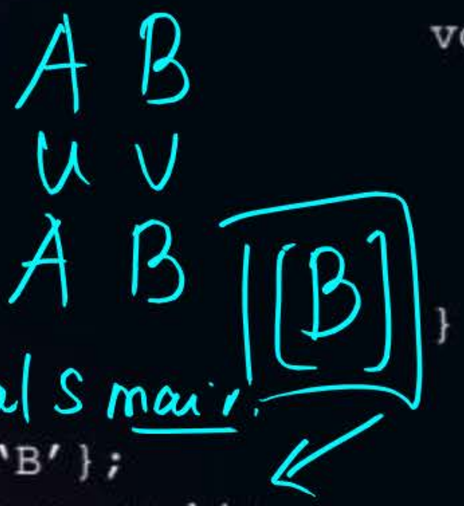
# GATE 2008

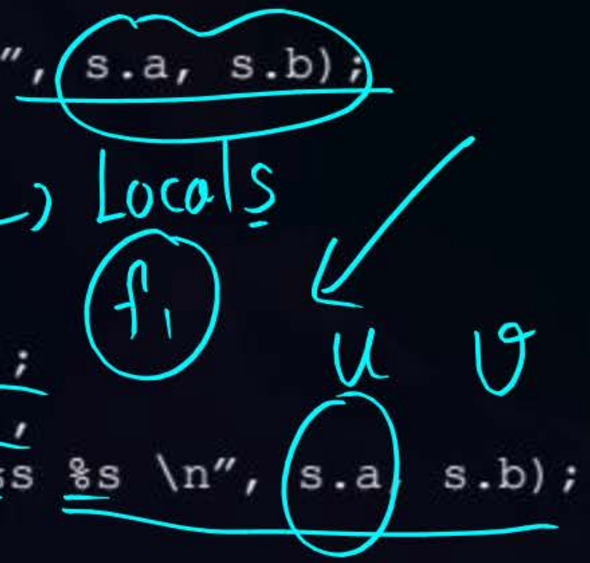
Consider the following C program:

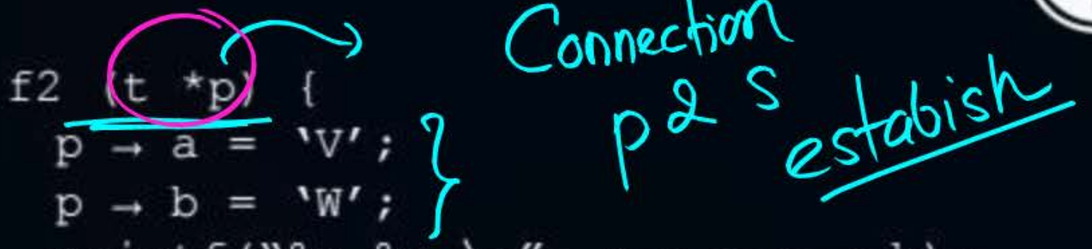
```
#include <stdio.h>
typedef struct {
        char *a;
        char *b;
} t;
void f1 (t s);
void f2 (t *p);
main(){
        static t s = {'A', 'B'};
        printf("%s %s \n", s.a, s.b);
        f1(s);
        printf("%s %s \n", s.a, s.b);
        f2(&s);
}

void f1 (t s){
        s.a = 'U';
        s.b = 'V';
        printf("%s %s \n", s.a, s.b);
        return;
}

void f2 (t *p) {
        p → a = 'V';
        p → b = 'W';
        printf("%s %s \n", p → a, p → b);
        return;
}
```




| | |
|---|---|
| (A) AB<br>UV<br>VW<br>VW | (B) AB<br>UV<br>AB<br>VW |
| (C) AB<br>UV<br>UV<br>VW | (D) AB<br>UV<br>VW<br>UV |

Slide



# Union

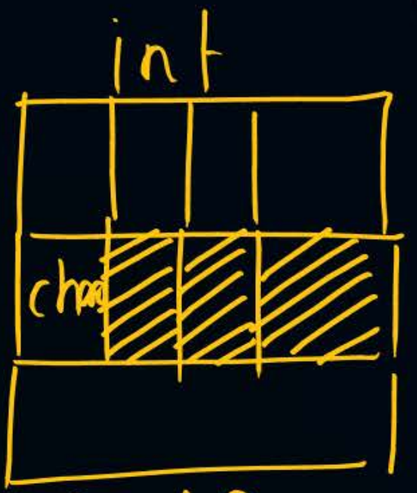
```
struct temp{
    int i;
    char ch;
};
```

4B

1B

5B

alingment constraints




4B+1B=5

New variable Locahon staot with value which divicke by 4

Slide



# Union

Every pointer (Any type)
is fixed Size = 4B

Slide



# Union

Union allows to use same more location to store different data type.

Set Theory union (OR)

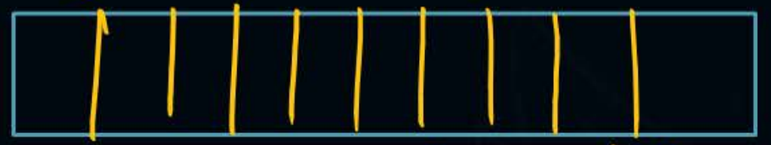


# Question

```
#include <stdio.h>
#include <string.h>
union Data{
   int i;
   float f;
   char  str[32];
};
int main(){
  union Data data;
  printf( "Memory size occupied by data : %d\n", sizeof(data));
  return 0;
}
```

4B
4B
32 bits

Location Size is equal to maximum memory required.




integer / float / string

keyword Sizeof

Slide

# Question

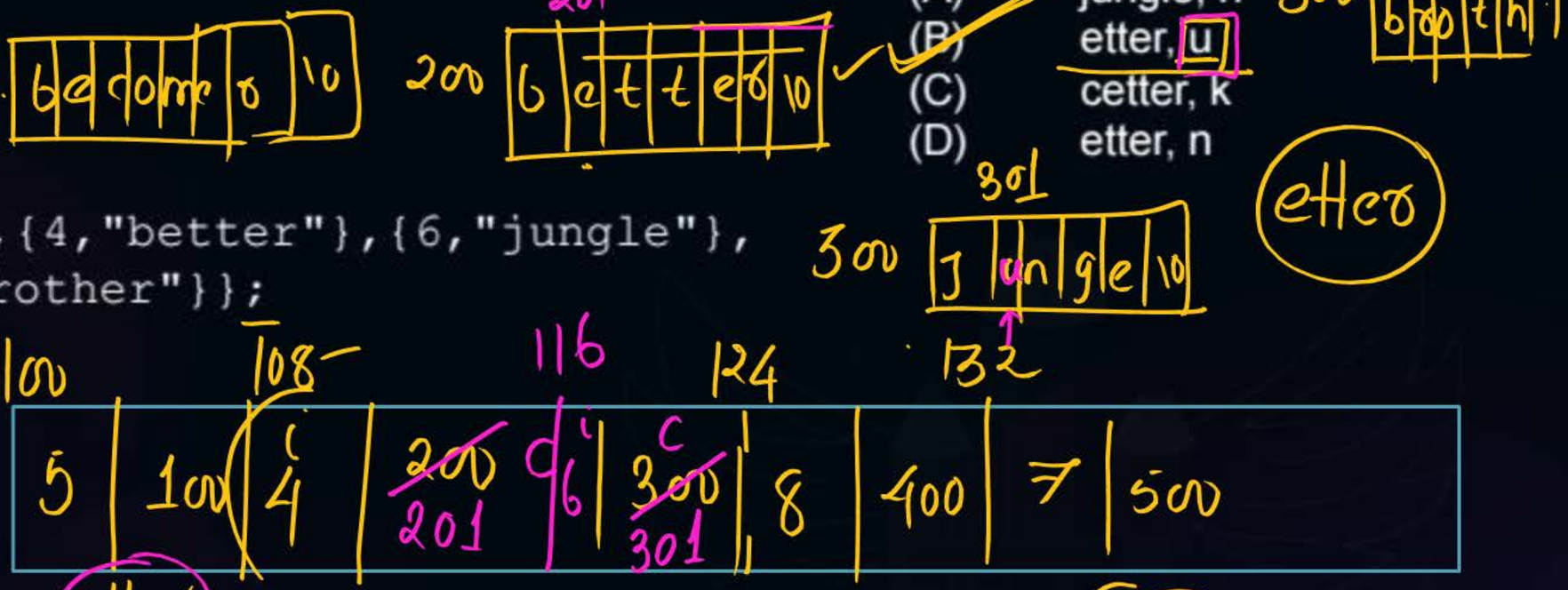
Which one of the choices given below would be printed when the following program is executed?

```
#include<stdio.h>
struct tes{
 int i;
char *c;
}st[5]={{5,"becomer"},{4,"better"},{6,"jungle"},
{8,"ancestor"},{7, "brother"}};

main (){
 struct tes *p=st;
 p+=1;
++(p->c); //
 printf("%s, ",p++->c);
 printf("%c",*++p->c);
}
```

*++(p→c)

(A) jungle, n
(B) etter, u
(C) cetter, k
(D) etter, n



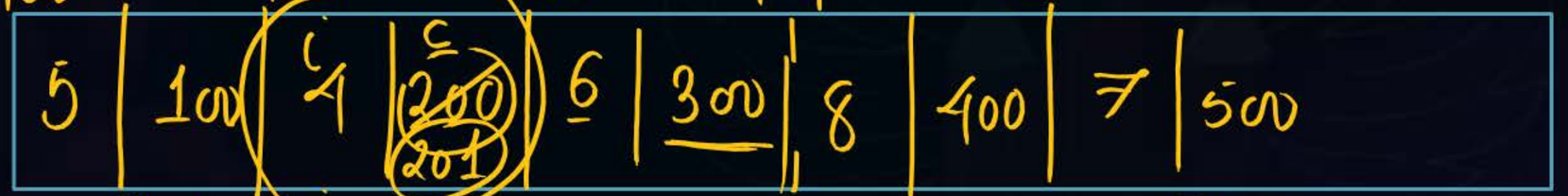
# Question

Which one of the choices given below would be printed when the following program is executed?

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main (){
 struct tes *p=st;
 p+=1;
 ++(p->c); //
 printf("%s, ",p++->c);
 printf("%c",*++p->c);
}
```

(A) jungle, n
(B) etter, u
(C) cetter, k
(D) etter, n




Slide



# #define Macro

A macro is a symbol that is recognized by the preprocessor and replaced by the macro body

Language processor

: preprocessor

#include<stdio.h>

macro expansion



# #define Macro

#define pi 3.14

macro

#define Max 100 ← No Semicolon

Example:

#define BUFFERSZ 1024

#define WORDLEN 64

program ⟶ if ( Size == Max ) ✓

preprocessor will replace each occurrence of max

100

if ( Size == 100 )

Slide



# #define Macro

#define pi 3.14

Example:

```
#define   BUFFERSZ   1024
#define   WORDLEN    64
```

macro

#define Max 100 ← No Semi colon

= Name for constant value.

Slide



# 2 mins Summary

Topic Structure

Topic Union

Topic

Topic

Topic

Slide

# THANK - YOU

